

SQL Server Automated Backup & SAN Replication Solution

Enterprise Technical Documentation & Operations Guide

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Target Environment: Microsoft SQL Server (Windows Server)

Script Executable: SQLSERVER_AUTO_BACKUP_EXTERNAL_HARDDRIVE.bat

1. Overview and Architecture

This enterprise-grade batch utility automates the complete end-to-end database backup, compression, external replication, and scheduling lifecycle for Microsoft SQL Server instances. Engineered to operate securely via manual execution or automated Windows Task Scheduler triggers, the script systematically mitigates common administrative failure points including missing permissions (Error 916), absent compression dependencies, and unhandled network share drops.

2. Configuration Parameters

The operational behavior and target infrastructure are governed by administrative configuration variables defined at the head of the batch script:

Parameter Variable	Description	Example Value
SERVER_NAME	SQL Server instance name or IP address endpoint.	localhost\SQLEXPRESS
DB_USER	SQL Server authentication username (blank for Windows Auth).	sa or BackupOperator
DB_PASS	Authentication password corresponding to the database user.	SecurePassword123
DB_NAME	Target database identifier slated for backup execution.	ProductionDB
BACKUP_DIR	Local storage staging directory path for raw backups.	C:\SQLBackups

Parameter Variable	Description	Example Value
<code>SAN_DRIVE</code>	Assigned drive letter designated for network share mapping.	<code>Z:</code>
<code>SAN_PATH</code>	Network UNC path pointing to the remote SAN/backup storage.	<code>\\IP_ADDRESS\Backups</code>
<code>SAN_USER / PASS</code>	Domain or network credentials required for remote share access.	<code>Domain\Credentials</code>
<code>EXTERNAL_DIR</code>	Destination target directory located on the external drive/SAN.	<code>Z:\</code>

3. Step-by-Step Execution Workflow

The automation script executes sequentially through six discrete operational phases:

Phase 1: Privilege Escalation & Self-Stabilization

Verifies elevated administrative privileges. If executed manually without elevation, it automatically spawns an elevated PowerShell process via `Start-Process -Verb RunAs` and clones itself into `\Scripts\` to ensure a stable execution path for scheduler tasks.

Phase 2: Database Security Provisioning

Provisions `NT AUTHORITY\SYSTEM` as a valid SQL Server login and assigns `sysadmin` server role permissions, preventing automated failure exceptions like Error 916.

Phase 3: Database Backup (.bak Generation)

Generates high-precision timestamps (`YYYY-MM-DD_HH-MM-SS`) via WMIC and invokes `sqlcmd` to execute a full database backup with formatted media settings.

Phase 4: Utility Detection & Compression (.rar)

Locates WinRAR (`Rar.exe`), automatically downloading and silently installing the binary over TLS 1.2 if absent. Compresses raw backups using maximum mode (`-m5`) and purges uncompressed data.

Phase 5: Network SAN / External Drive Replication

Clears stale network mappings on `SAN_DRIVE` to avert conflict error 85, mounts the remote UNC path via `net use` credentials, provisions target directories if missing, and replicates the archive.

Phase 6: Windows Task Scheduler Registration

Registers or updates a persistent daily scheduled task running at 8:00 AM under `NT AUTHORITY\SYSTEM` with highest privileges, injecting advanced resiliency settings (battery execution allowances, overlap ignores, and auto-retries).

4. Output & Log Analysis

Execution telemetry streams concurrently to the active console and `backup_log.txt`. Below is a breakdown of operational outputs:

```
== Execution Started: 09/29/2026 11:00:00 AM Target External Directory: Z:\ Granting NT AUTHORITY\SYSTEM access to SQL Server... [SUCCESS] STEP 1: Running immediate backup for database 'ProductionDB'... [SUCCESS] STEP 2: Checking for WinRAR / Rar.exe utility... [SUCCESS] STEP 3: Compressing bak file into .rar (Mode -m5)... [SUCCESS] STEP 4: Authenticating and Copying archive to SAN / External Drive... [SUCCESS] STEP 5 & 6: Task Registration Confirmed (Next Run: 9/30/2026 8:00 AM, Status: Ready) Process completed! Check log at: C:\SQLBackups\backup_log.txt
```

The runtime concludes within approximately 15 seconds, securing fully audited database snapshots off-box while maintaining complete state integrity across system reboots.